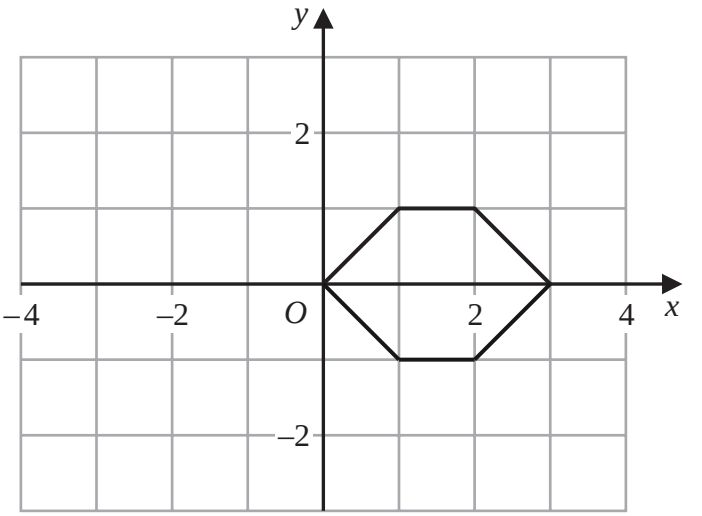
18 The graph of $y = f(x)$ is shown on the grid.



(a) On the grid, draw the graph with equation $y = -f(x)$

1 mark

Reflect every original point in the x-axis.

(1)

The curve with equation $y = 2x^2 + 5x - 3$ is transformed to the curve **T** by a translation with vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation of the curve **T**.
Give your answer in the form $y = ax^2 + bx + c$ where a, b and c are integers.

Translation4 units right means replace x by (x –4).

$y = 2(x-4)^2 + 5(x-4) - 3$

$= 2(x^2 - 8x + 16) + 5x - 20 - 3$

$= 2x^2 - 16x + 32 + 5x - 20 - 3$

$= 2x^2 - 11x + 9$

1 mark

Use f(x –4), then expand the squared bracket.

$2x^2 - 11x + 9$

(2)

(Total for Question 18 is 3 marks)

1 mark

Collect like terms; the vertical translation is zero.

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